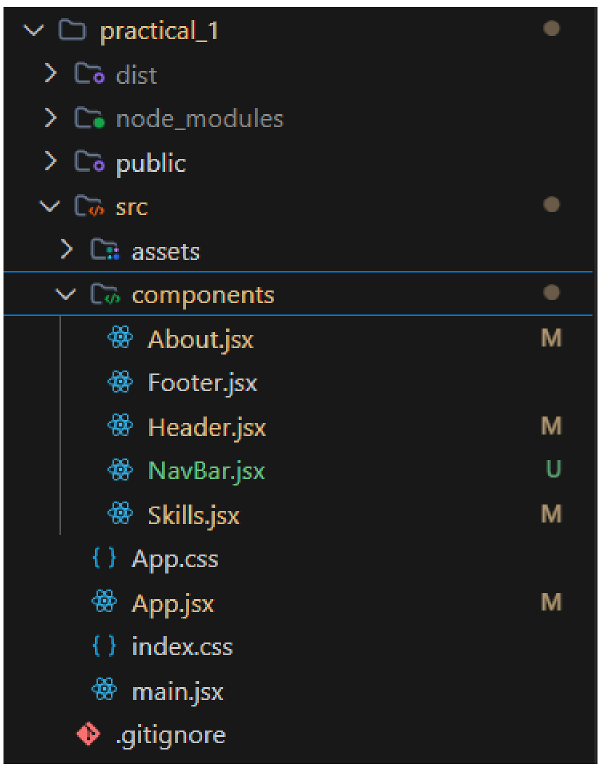
Practical 1: Introduction to React and Component Architecture

Problem Statement

• Create a React application using Vite for a student portfolio page.
• The application must include at least 4 reusable components:
Header, About, Skills, and Footer.
• Each component must be independently structured and composed
into a single-page layout.
• No logic or JSX should be duplicated across components.
• Props must be used to pass at least one piece of data into a
minimum of 2 components.

Project Structure:



Source Code Screenshots:

main.jsx

```jsx
import { StrictMode } from 'react'
import { createRoot } from 'react-dom/client'
import './index.css'
import App from './App.jsx'
import "./App.css";

createRoot(document.getElementById('root')).render(
  <StrictMode>
    <App />
  </StrictMode>,
)
```

App.jsx

```jsx
practical_1 > src > App.jsx > ...
import React, { useState, useEffect } from "react";
import Header from "./components/Header";
import NavBar from "./components/NavBar";
import About from "./components/About";
import Skills from "./components/Skills";
import Footer from "./components/Footer";

function App() {
  const studentName = "Rudra Joshi";
  const [activeSection, setActiveSection] = useState("about");
  const themeColor = "#2e7d32"; // Dark green theme color for inline style demo

  useEffect(() => {
    const params = new URLSearchParams(window.location.search);
    const section = params.get("section");
    if (section) {
      setActiveSection(section);
      setTimeout(() => {
        const element = document.getElementById(section);
        if (element) {
          element.scrollIntoView({ behavior: "auto", block: "start" });
        }
      }, 100);
    }
  }, []);

  const skillList = [
    "React JS",
    "JavaScript",
    "HTML",
    "CSS",
    "Node JS"
  ];

  return (
    <div className="container">

      <NavBar activeSection={activeSection} onSectionClick={setActiveSection} />

      <Header name={studentName} themeColor={themeColor} />

      <About />

      <Skills skills={skillList} />

      <Footer name={studentName} />

    </div>
  );
}

export default App;
```

Header.jsx

```
function Header(props) {

    return (
        <header className="header">

            <h1>Student Portfolio</h1>

            <h2>{props.name}</h2>

        </header>
    );
}

export default Header;
```

About.jsx

```
function About() {

    return (

        <section className="about">

            <h2>About Me</h2>

            <p>
                Hello, I am a Information Technology Student.
                I love creating web applications using React.
            </p>

        </section>

    );
}

export default About;
```

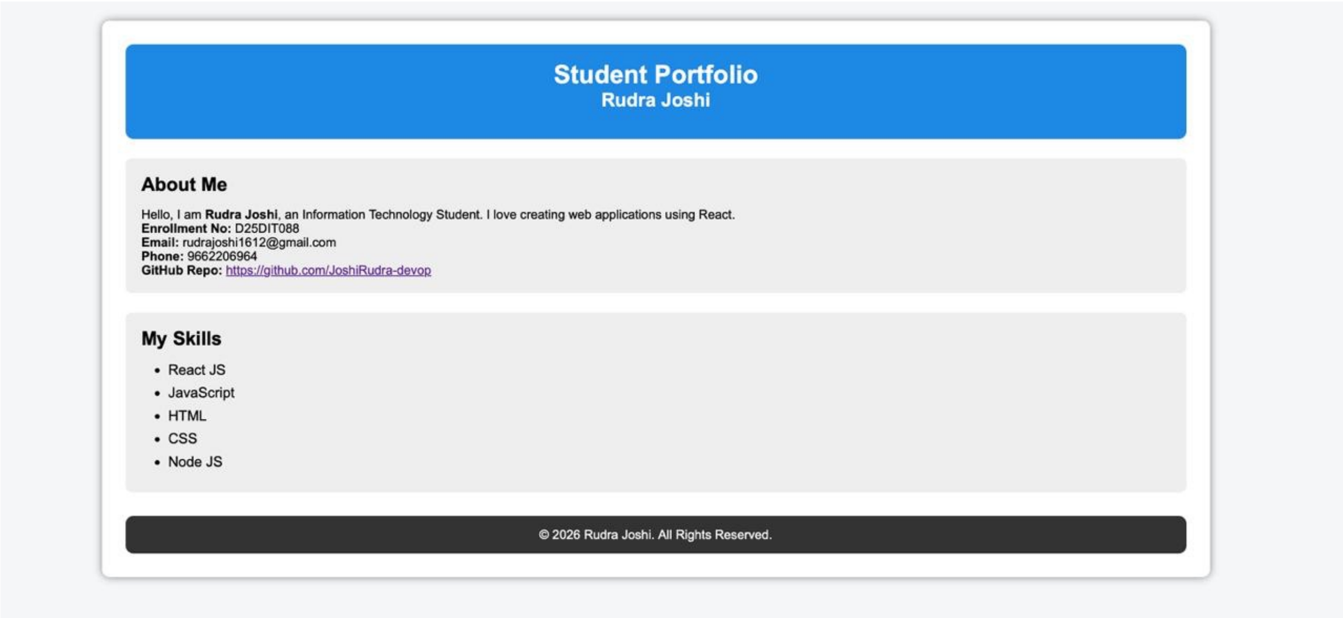
Skills.jsx

```
function Skills(props) {

    return (

        <section className="skills">

            <h2>My Skills</h2>

            <ul>

                {
                    props.skills.map((skill, index) => (
                        <li key={index}>
                            {skill}
                        </li>

                    ))
                }

            </ul>

        </section>

    );
}

export default Skills;
```

Footer.jsx

```
function Footer(props) {

    return (

        <footer className="footer">

            <p>
                © 2026 {props.name}. All Rights Reserved.
            </p>

        </footer>

    );
}

export default Footer;
```

Output Screenshot:



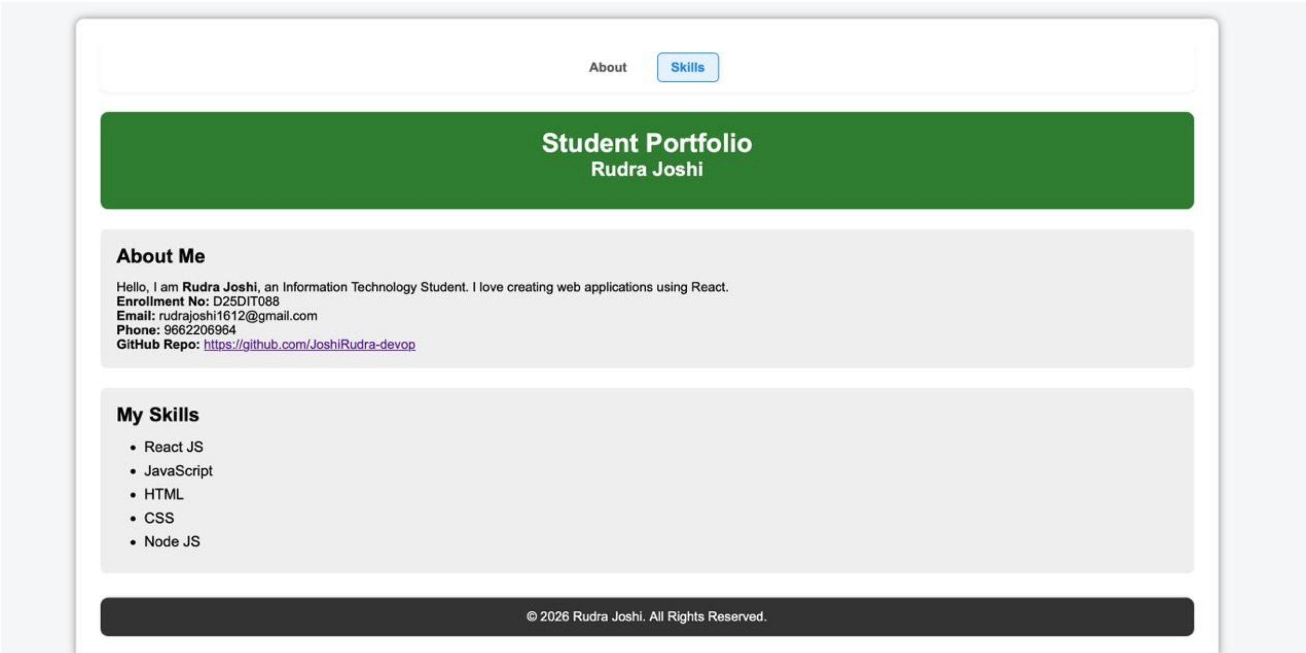
Supplementary Problems

1. NavBar Component: Created a new NavBar.jsx component displaying navigation links that update the active state and smoothly scroll to the selected section on click.
2. Dynamic Skills: Configured the Skills.jsx component to accept skills as an array prop and dynamically render them as list items.
3. Theme Color Prop: Passed a themeColor prop to Header.jsx and applied it inline as style={{ backgroundColor: props.themeColor }} to set a custom green color theme.

Source Code of NavBar.jsx

```
import React from "react";

function NavBar({ activeSection, onSectionClick }) {
  const sections = ["About", "Skills"];
  return (
    <nav className="navbar" style={{
      display: "flex", justifyContent: "center", gap: "20px", marginBottom: "25px",
      padding: "12px 20px", background: "#ffffff", borderRadius: "8px", boxShadow: "0 2px 5px
rgba(0,0,0,0.05)"
    }}>
      {sections.map((section) => {
        const isSelected = activeSection === section.toLowerCase();
        return (
          <a
            key={section}
            href={`#${section.toLowerCase()}`}
            style={{
              textDecoration: "none", color: isSelected ? "#1e88e5" : "#555",
              fontWeight: "600", fontSize: "16px", padding: "8px 16px", borderRadius: "6px",
              background: isSelected ? "#e3f2fd" : "transparent", transition: "all 0.3s ease",
              border: isSelected ? "1px solid #1e88e5" : "1px solid transparent"
            }}
            onClick={(e) => {
              e.preventDefault();
              onSectionClick(section.toLowerCase());
              const element = document.getElementById(section.toLowerCase());
              if (element) {
                element.scrollIntoView({ behavior: "smooth", block: "start" });
              }
            }}
          >
            {section}
          </a>
        );
      })}
    </nav>
  );
}
```

Output Screenshots of Supplementary Problems



Portfolio page showing the active section navbar highlighting 'Skills' and the header styling set dynamically via a theme color prop.